

Saira Rizvi

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RESEARCH INTERESTS

AI risk assessment and mitigation; trustworthy and safe AI systems; misinformation and information integrity; adversarial robustness of machine learning systems; applied machine learning for security and risk domains.

EDUCATION

University of Waterloo — Master of Mathematics, Data Science 2019 – 2021
Cumulative GPA: 85.80/100. Thesis: “Misinformation Retrieval” [\[link\]](#), supervised by Prof. Charles L. A. Clarke.

University of Toronto — Honours Bachelor of Science, Mathematics & Statistics 2015 – 2019
Conferred with Distinction. Cumulative GPA: 3.42/4.0. Achieved a 4.0/4.0 sessional GPA in third year (Winter 2018). Recipient of an undergraduate scholarship in third year.

PUBLICATIONS

- Clarke, C. L. A., Maistro, M., Rizvi, S., Smucker, M. D., & Zucco, G. (2020). Overview of the TREC 2020 Health Misinformation Track. In Proceedings of the Twenty-Ninth Text REtrieval Conference (TREC 2020), National Institute of Standards and Technology (NIST) [\[link\]](#).

Served as a track coordinator for the TREC Health Misinformation Track, contributing to topic development, document collection curation, and evaluation methodology for a shared task addressing information retrieval in the presence of health misinformation.

PATENTS

- Systems & Methods for Phishing Campaign Detection, US20250337778A1 (Filed 2025) [\[link\]](#)
- Detection of Phishing Websites Using Machine Learning, CA3170593A1 (Filed 2022) [\[link\]](#)

RESEARCH & APPLIED EXPERIENCE

RBC — Security Insights & AI | Staff Data Scientist & Manager May 2024 – Present
Applied ML research and production systems for fraud and behavioral threat detection

- Built and deployed a detection system for coordinated political advocacy and harassment email campaigns targeting senior executives, modeling attacker-to-target structure as a complementary signal to content-based classification and catching campaigns missed by existing security guardrails.
- Led development of a fraud detection model projected to capture \$10M CAD in fraud annually, from problem formulation through production deployment and evaluation.
- Designed a behavioral modeling approach for insider threat detection, establishing which risk signals hold discriminative power under strict false-positive constraints and where the approach breaks down.
- Led a cross-team research effort on spear-phishing detection, quantifying coverage gaps in existing defences and defining the team’s future directions.

RBC — Security Insights & AI | Senior Data Scientist Sept 2021 – May 2024
Anomaly detection for phishing and email-based threats

- Designed and deployed a phishing campaign detection system using anomaly detection, becoming the top-performing model on the team.
- Increased email threat detection coverage from ~20% to ~60% bank-wide through iterative feature and model research.
- Translated applied research into patented phishing detection methods.
- Built the biometric classification prototype that established keystroke dynamics as a viable identity-based risk signal, informing the team's subsequent detection roadmap.

University of Waterloo | Master’s Thesis & TREC Research Sept 2019 – Sept 2021
Misinformation retrieval and detection

- Co-authored the TREC 2020 Health Misinformation Track overview and authored the topic-selection and adjudication protocol behind its labels; the test collection was adopted in later research.
- Built a COVID-19 misinformation detector (BERT/NLI); diagnosed a train/deploy register mismatch and recovered lost accuracy via continued pretraining on in-domain text.

MENTORING

- Mentored junior data scientists at RBC on research scoping, experimental design, and evaluation practices.

HONORS & AWARDS

- RBC “Drive to Impact” Yearly Award (2026)
- RBC Individual Contribution Quarterly Award (2026) – for high impact contributions over years in Global Cybersecurity Division at RBC
- RBC Performance Recognition (2025)
- RBC Team Performance Award (2024)
- 3rd Place, Best Cybersecurity Patent, RBC (2024)
- Honours B.Sc. Conferred with Distinction, University of Toronto (2019)

TECHNICAL SKILLS

- Languages & Frameworks: Python, SQL, PySpark, Scikit-learn, PyTorch, Hugging Face
- Methods: Machine Learning, Anomaly detection, behavioral modeling, BERT/transformer fine-tuning, information retrieval, statistical modeling
- Infrastructure: AWS, Git, Jupyter, production ML pipelines

REFERENCES

Available on request